

Introduction to Next Generation Sequencing

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Agenda for the next weeks

Date	Topic
24.09.2026	Introduction to NGS and conda
25.09.2026	Data Quality Control, Trimming and Read Mapping
28.09.2026	
29.09.2026	Variant calling and annotation
30.09.2026	
01.10.2026	RNA-Seq data analysis (in R) - Mapping and Quantification of gene expression
02.10.2026	Detecting differential gene expression (normalisation, statistical testing, visualisation)
05.10.2026	Workflow management with Nextflow
06.10.2026	Workflow management with Nextflow
07.10.2026	Single-cell RNA-Seq

Goals

- Understanding the NGS vocabulary and the concepts behind the most common analysis workflows
- Gather experience in handling NGS data:
 - installing new software tools, understanding how to run them
 - integration in an analysis pipeline
 - troubleshooting

Agenda for today

- Brief history of sequencing / NGS Applications
- Our focus: Illumina sequencing technology
- NGS Data: Reads, Data Formats and Quality metrics
- Practical introduction to conda

Brief history of sequencing

First generation sequencing (1970's)

Sanger Sequencing

Second generation sequencing (2000's)

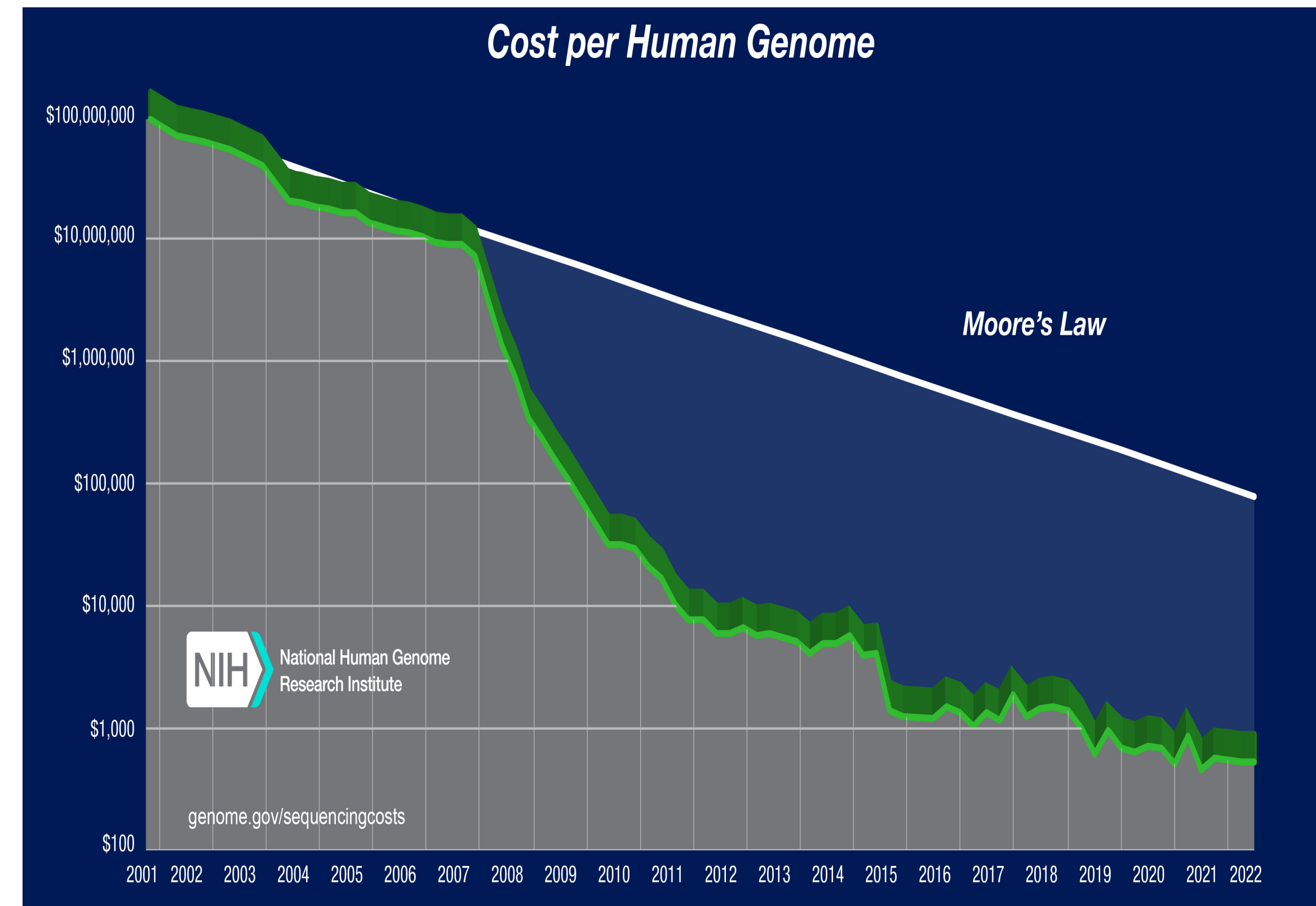
pyrosequencing (454 Life Sciences)

sequencing-by-ligation (ABI SOLiD)

sequencing-by-synthesis (Illumina)

Third generation sequencing (~2008-2009)

long-read sequencing (PacBio, Oxford Nanopore)



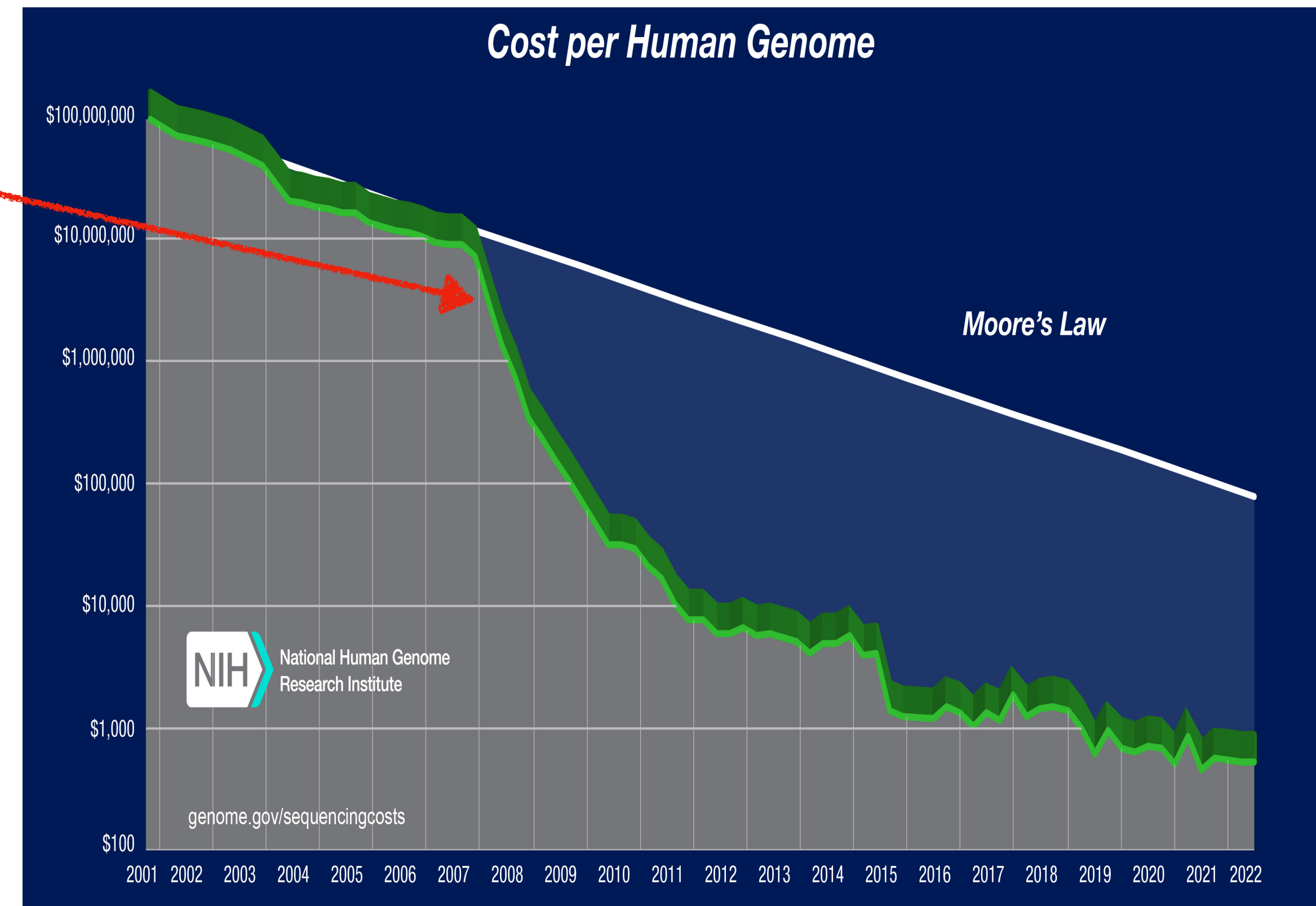
Brief history of sequencing

What has happened here?

- Solexa was acquired by Illumina
- Sequencing costs have dropped at a rate that outpaces Moore's law
- NGS data output has increased at a rate that outpaces Moore's law

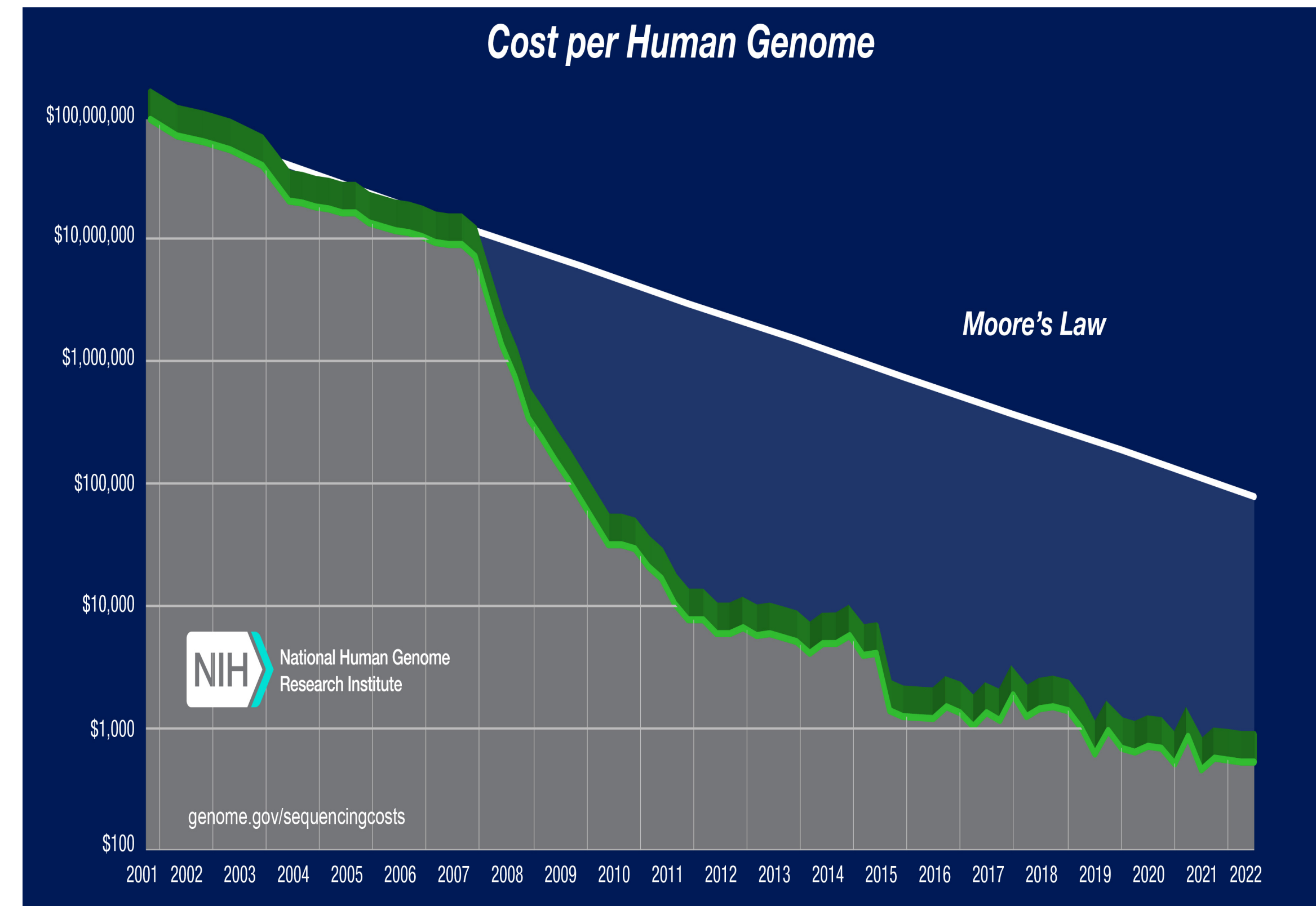
Moore's Law:

The number of transistors in an integrated circuit doubles about every two years.

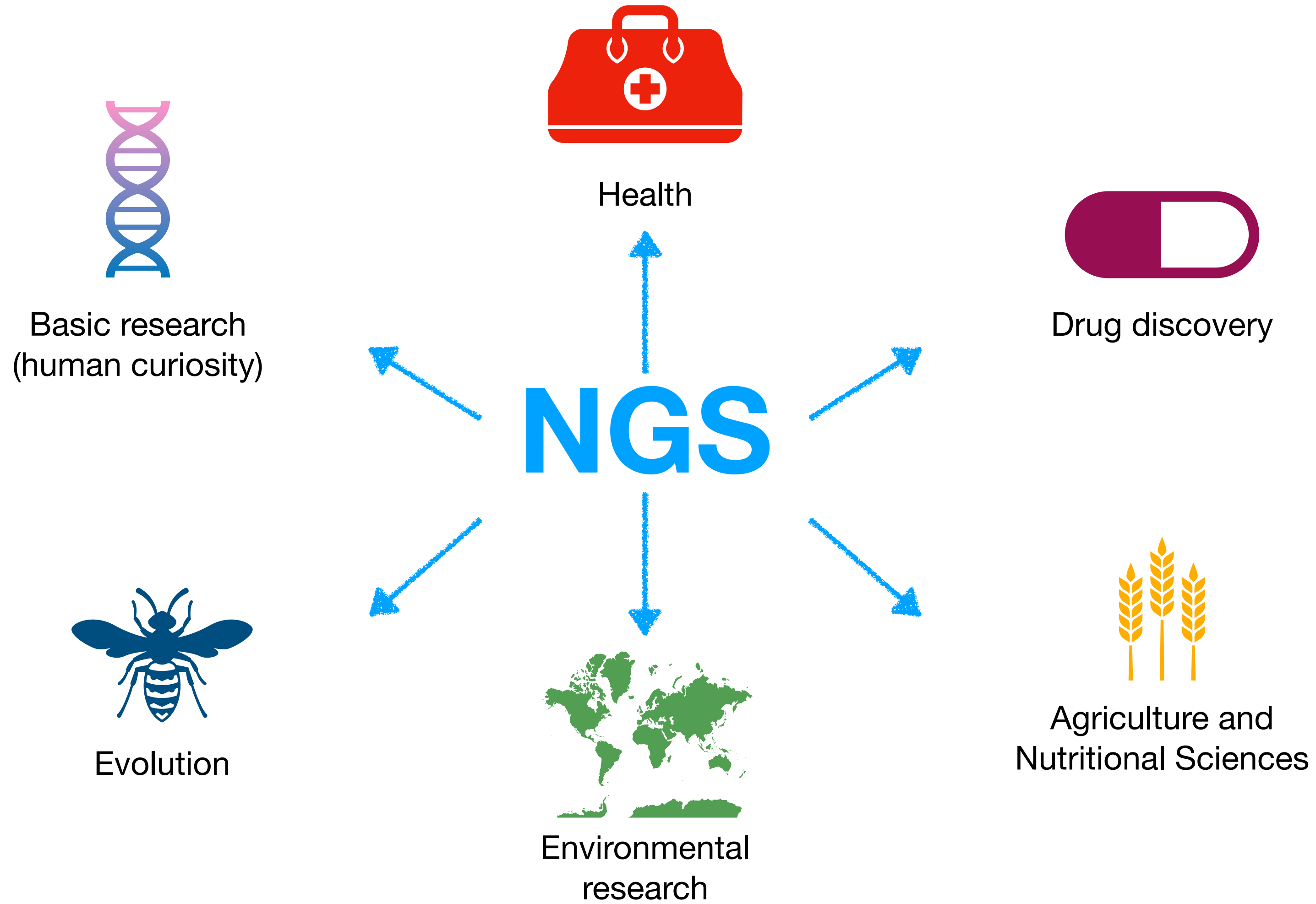


Trends

- **New players** -> “100\$ genome”
 - Ultima Genomics
 - Element Biosciences
 - Roche
- **Environmental awareness**
 - Reusability
 - Less reagents at room temperature
 - Less packaging



Applications



NGS in Health

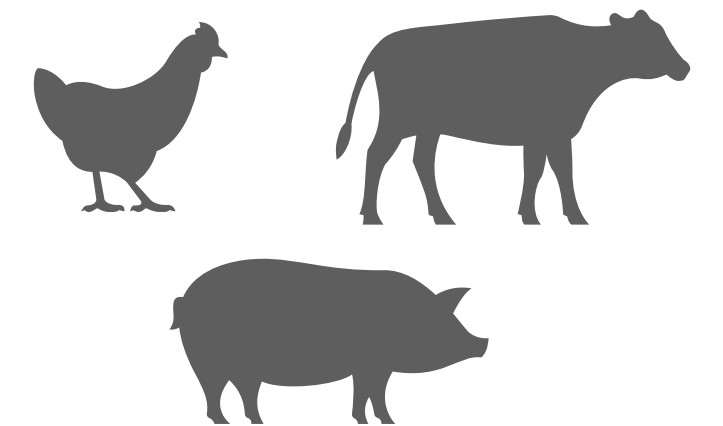
- **whole-genome sequencing (WGS) and targeted sequences** (whole-exome, panels, or amplicons) widely applied in **diagnostics of rare diseases and cancer**
- **16S-, ITS-, whole-genome metagenomics** give insights into the **role of the human microbiome on our health**
- **the era of personalised medicine** through pharmacogenomics, genetic risk scores, and immunogenomics

NGS in Drug Discovery and Cell Therapy

- **Immunogenomics:** sequencing the variable, diversity and joining loci of the B-cell receptors and T-cell receptors to characterise the immune profile.
- Physiological effect of treatments through **RNA-Seq**
- Vaccine development
- NGS is being applied also in cosmetics

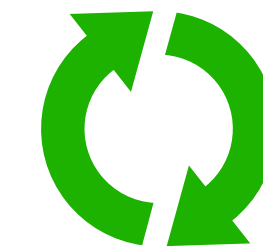
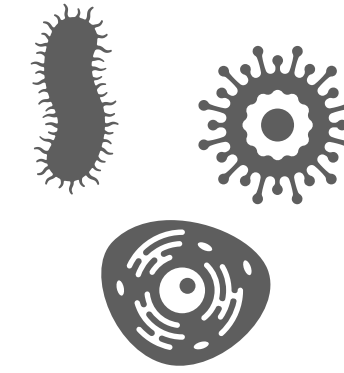
NGS in Agricultural Sciences

- identification of **genetic markers** associated with traits of interest:
 - drought resistance, pathogen resistance, yield increase
- animal breeders search for positive genotypes for **economically important qualities**
- food control, testing for contamination



NGS in Environmental Sciences

- Metagenomics of soil, water, stool samples etc.
- **Taxonomic profiling** up to species level
- Identification of **pathogens**
- Identification of **antibiotics resistance genes**
- **Biodiversity**

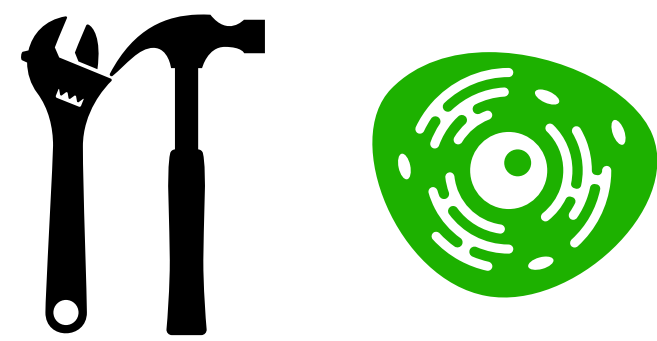


NGS in Evolution Research

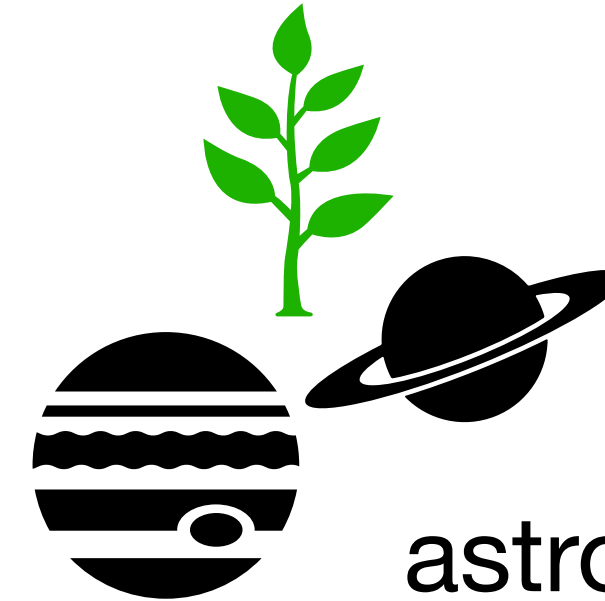
- Taxonomy and Species differentiation
- Phylogenetics: Please check out the **Interactive Tree of Life** (<https://itol.embl.de>)
- Ancestry (23andMe, etc.)

NGS in Basic Research

Goal: understanding molecular mechanisms of anything and everything



synthetic biology



astrobiology

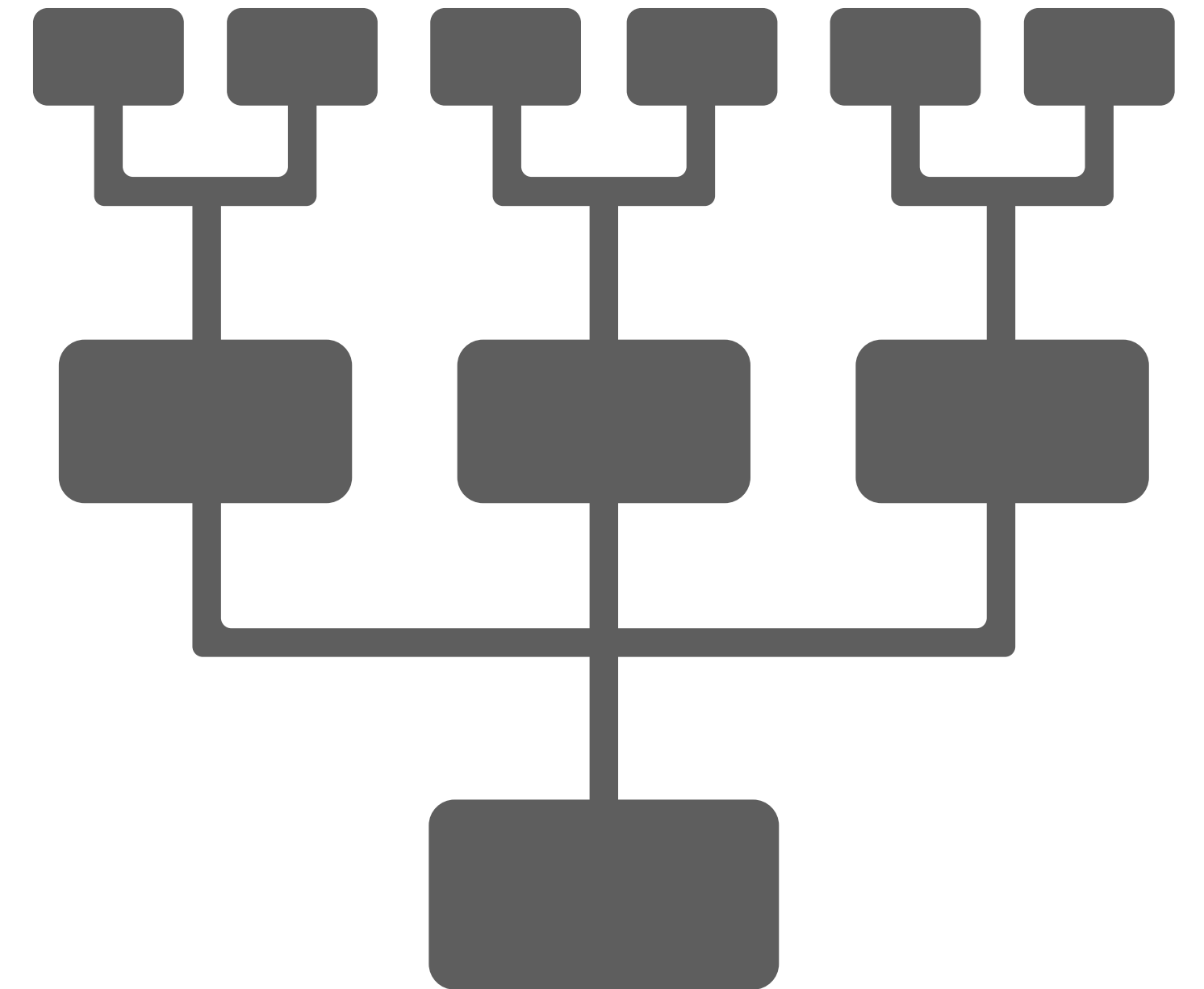


ageing and
increasing the lifespan

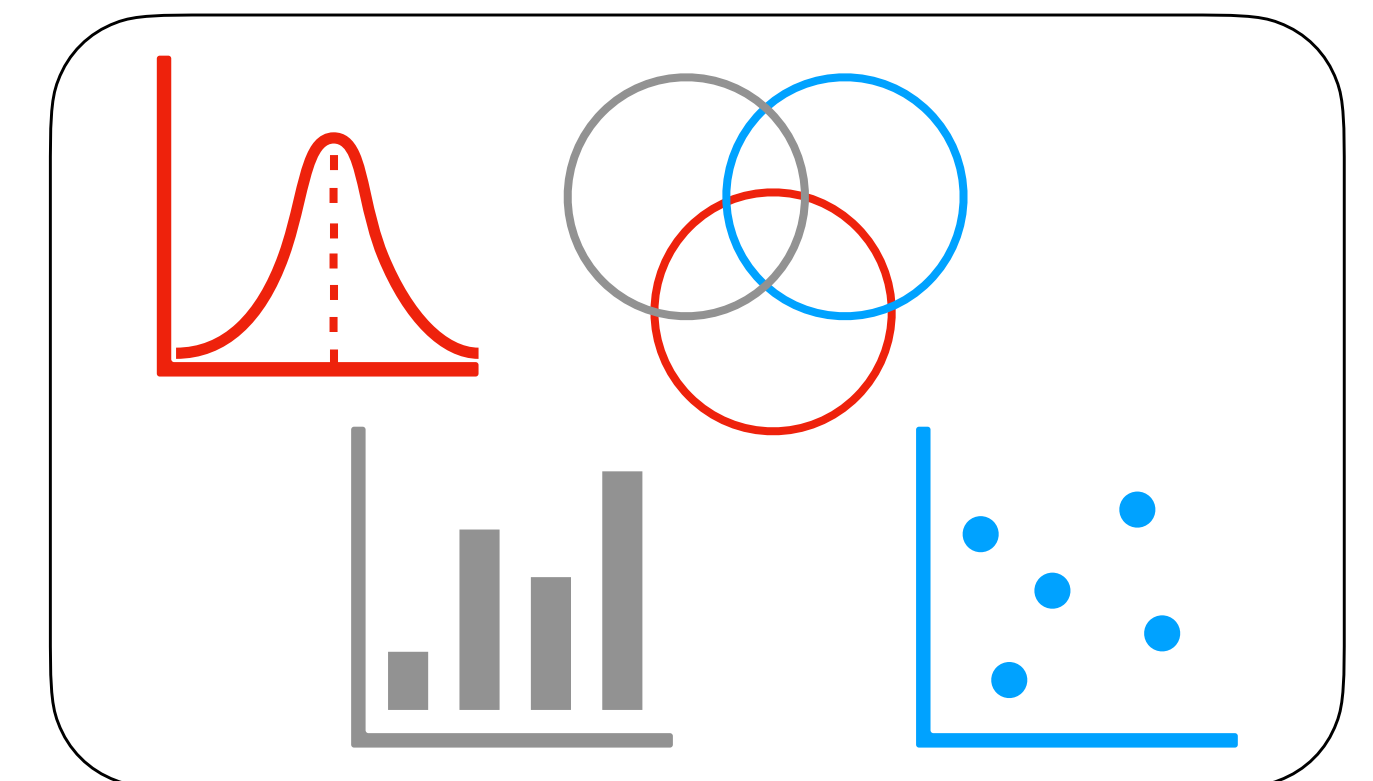
Next Generation Sequencing Data

- High-throughput / Big Data
- Illumina's latest sequencer NovaSeq X Plus can generate in a single run 16TB
- Bioinformatics is needed to handle the data and perform the data analysis
- Understanding the steps of the analysis and setting up the analysis pipeline while considering the available infrastructure is essential.

Raw data



Results



Illumina Sequencing Workflow

Sequencing by synthesis

Steps

1. Library preparation
2. Cluster generation
3. Sequencing
4. Demultiplexing and further data analysis

Great introductory video from Illumina: <https://www.youtube.com/watch?v=olJaA6h2bFM>

Library Preparation

- Starting material: genomic DNA
- DNA fragmentation (mechanical/ enzymatic) -> fragment size 250-350bp
- End conversion and adapter ligation
- PCR Amplification (optional, risk of overamplification)

Library Structure - Paired end sequencing



- **P7 and P5 binding regions**, which interact with oligos on the surface of the flow cell (clustering)
- **Unique index combination of i7 and i5** for mixing samples on the same flow cell (multiplexing)
- **Binding regions for sequencing primers** that enable the synthesis of R1 and R2 (sequencing)

Cluster generation

- A flow cell is a glass slide with fluidic channels or lanes, where the sequencing chemistry occurs
- Libraries are hybridised onto a flow cell followed by bridge amplification
- Result: distinct clonal cluster for each fragment on the flow cell

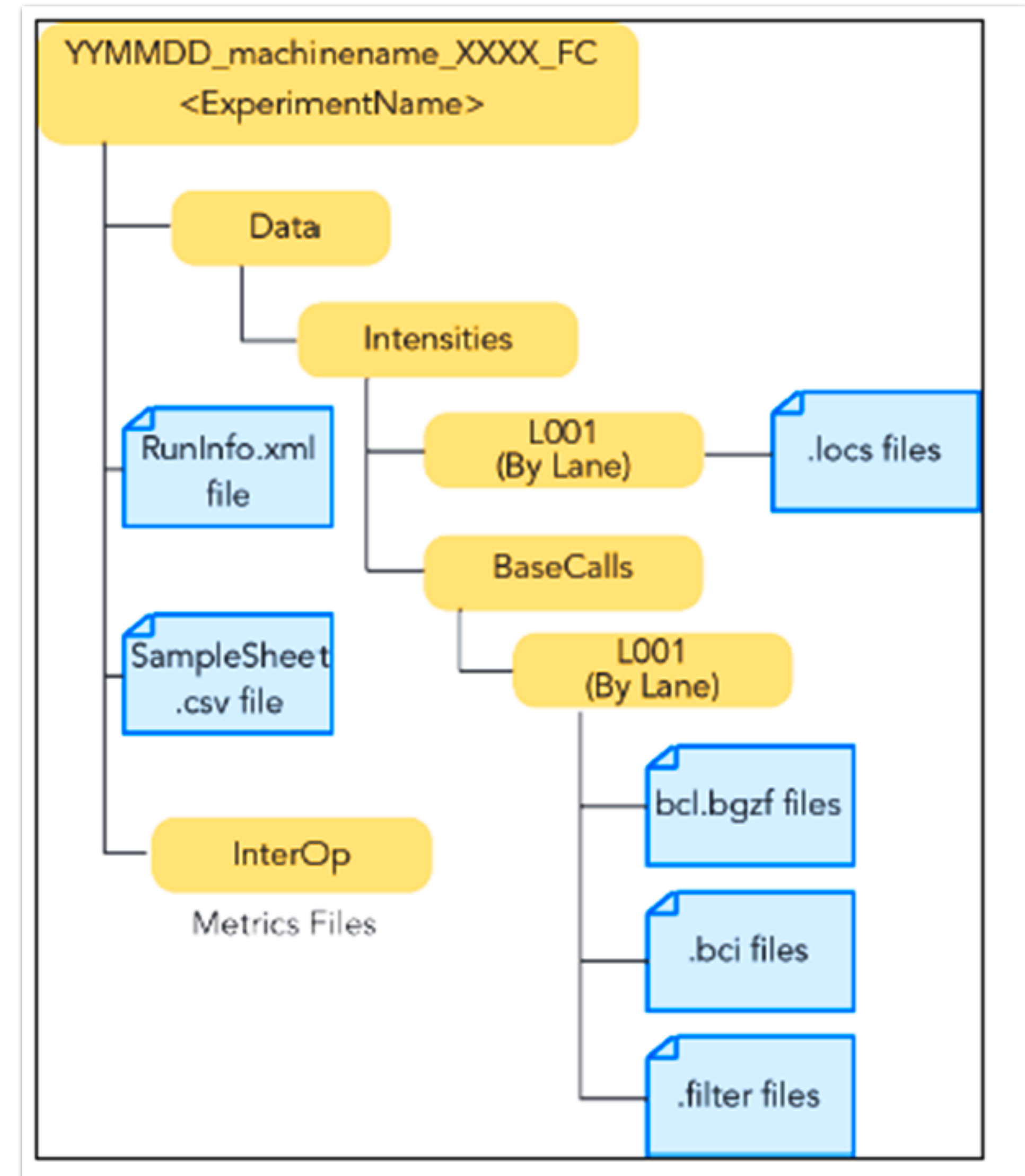
Sequencing

- Fluorescently labeled nucleotides are added that bind to the complementary bases in the cluster
- The sequencer images the flow cell for each cycle
- The image information identifies the base in the cluster (base position in read corresponds to cycle number)
- 4-Channel Chemistry / 2-Channel Chemistry

Raw run data

Run folder has a **well defined structure** and contains:

- Binary base call files (*.bcl, *.cbcl)
- Filter files (*.filter)
- Position files (*.locs, *.clocs, *.s.locs)
- Aggregated files (*.bci)
- Run info file (*.xml)
- to be added: sample sheet



Demultiplexing

- Demultiplexing is the step of assigning each read to a specific sample and generating fastq files
- This step is performed once the sequencing is completed
- Depending on the sequencer and its output size, the demultiplexing can run on the sequencer or on a separate platform (server, high-performance cluster, cloud)
- Software: **bcl2fastq2** or **BCL Convert**
- Extremely careful when mapping the library structure to the input parameters

FastQ file format

- For each sample two fastq files are generated: R1 and R2
- Typical file size: several GB **Note: Don't try to open them in Word or notepad!**
- Typical file extensions: .fq, .fastq, .fastq.gz (if compressed with gzip)
- Each entry consists of a 4 line-block:
 - sequence identifier
 - nucleotide sequence
 - quality score identifier line
 - quality score

https://support.illumina.com/help/BaseSpace_OLH_009008/Content/Source/Informatics/BS/FileFormat_FASTQ-files_swBS.htm

Fastq Format

Sequence identifier contains the following information:

```
@<instrument>:<run number>:<flowcell ID>:<lane>:<tile>:<x-pos>:  
<y-pos>:<UMI> <read>:<is filtered>:<control number>:<index>
```

Sequence identifier contains the following information:

```
@A01526:19:HFN7FDRX2:1:2101:4173:1016 1:N:0:GTTGGCGTCT+CTAGACTTCG  
GNTGTTTGAAATTATCACATATCATATAACCTCTTGAAAGGTACACTGTACACTTGTGAAAGAGTGA...  
+  
F#: ,F:FFFFFFFFFFFFFF:FFFFFF:FFFFF:FFFFF:FF,FFF:FFFFFF:,FF:F:FFFFFFFF:FF:...
```

Quality Score

- Phred Score (Q) = logarithmically related to the probability that the corresponding base call is incorrect

$$Q = -10 \log_{10}(P)$$

Q Score	Probability of incorrect base call	Base call accuracy
10	1 in 10	90 %
20	1 in 100	99 %
30	1 in 1000	99.9%
40	1 in 10,000	99.99%

Q30 metric = % of bases in a file having Q >=30

Quality Score

- Phred Score encoded as the ASCII sign corresponding to Q+33

```
@A01526:19:HFN7FDRX2:1:2101:4173:1016 1:N:0:GTTGGCGTCT+CTAGACTTCG
GNTGTTTGAAATTATCACATATCATATAACCTCTTGAAAGGTACACTGTACACTTGTGAAAGAGTGA...
+
F#: ,F:FFFFFFFFFFFFFF:FFFFFF:FFFFF:FFFFF:FF,FFF:FFFFFF:,FF:F:FFFFFFFF:FF:...
```

- Let us find out how the quality in this read looks like!

Quality Score

- Phred Score encoded as the ASCII sign corresponding to $Q+33$

```
@A01526:19:HFN7FDRX2:1:2101:4173:1016 1:N:0:GTTGGCGTCT+CTAGACTTCG
GNTGTTTGAAATTATCACATATCATATAACCTCTTGAAAGGTACACTGTACACTTGTGAAAGAGTGA...
+
F#: ,F:FFFFFFFFFFFFFF:FFFFFF:FFFFF:FFFFF:FF,FFF:FFFFFF:,FF:F:FFFFFFFF:FF:...
```

Q=2

Q=37

Q=25

Q=11

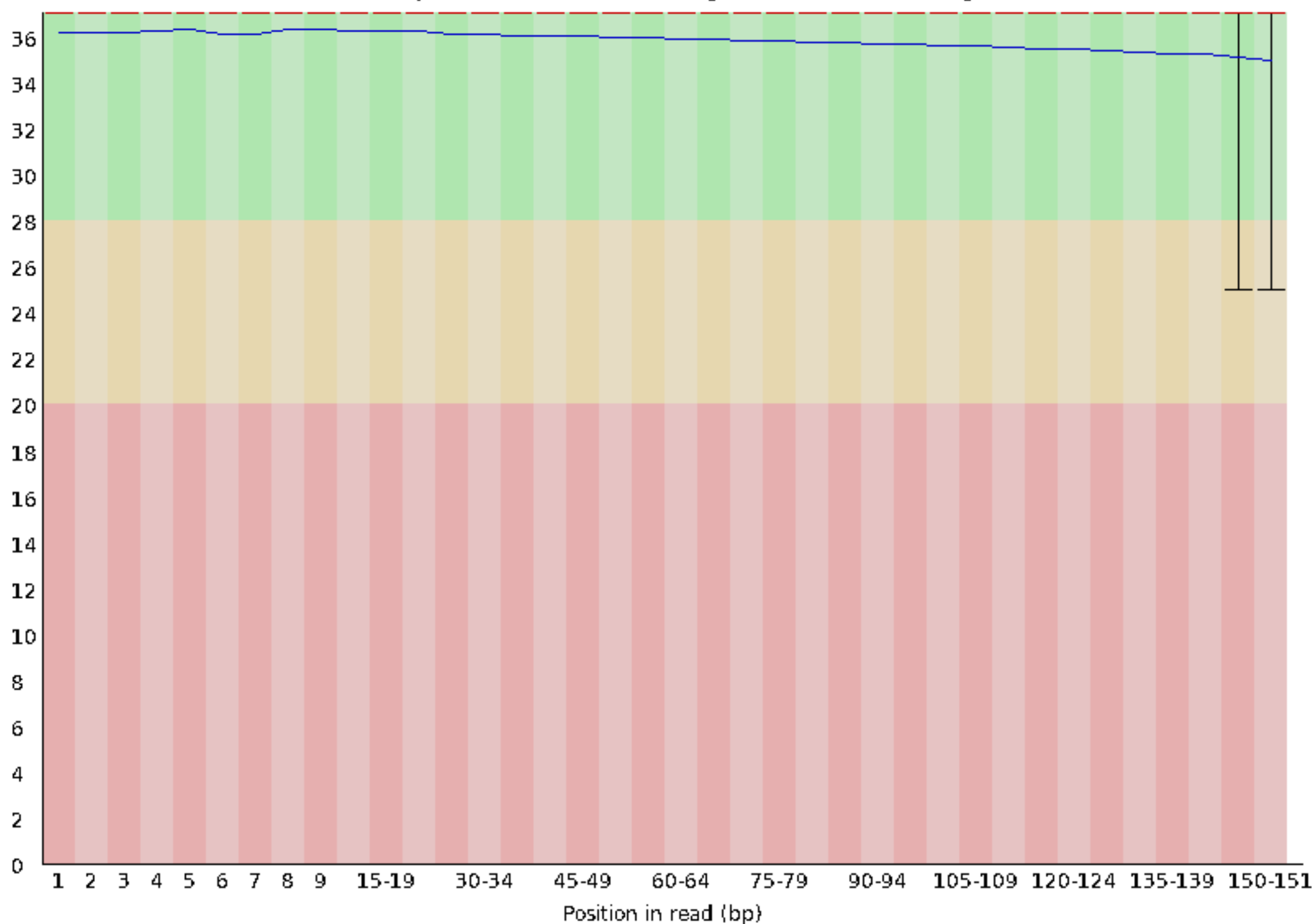
Quality control

FastQC - the standard tool

- <https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>
- well documented
- continuously maintained
- generates html reports

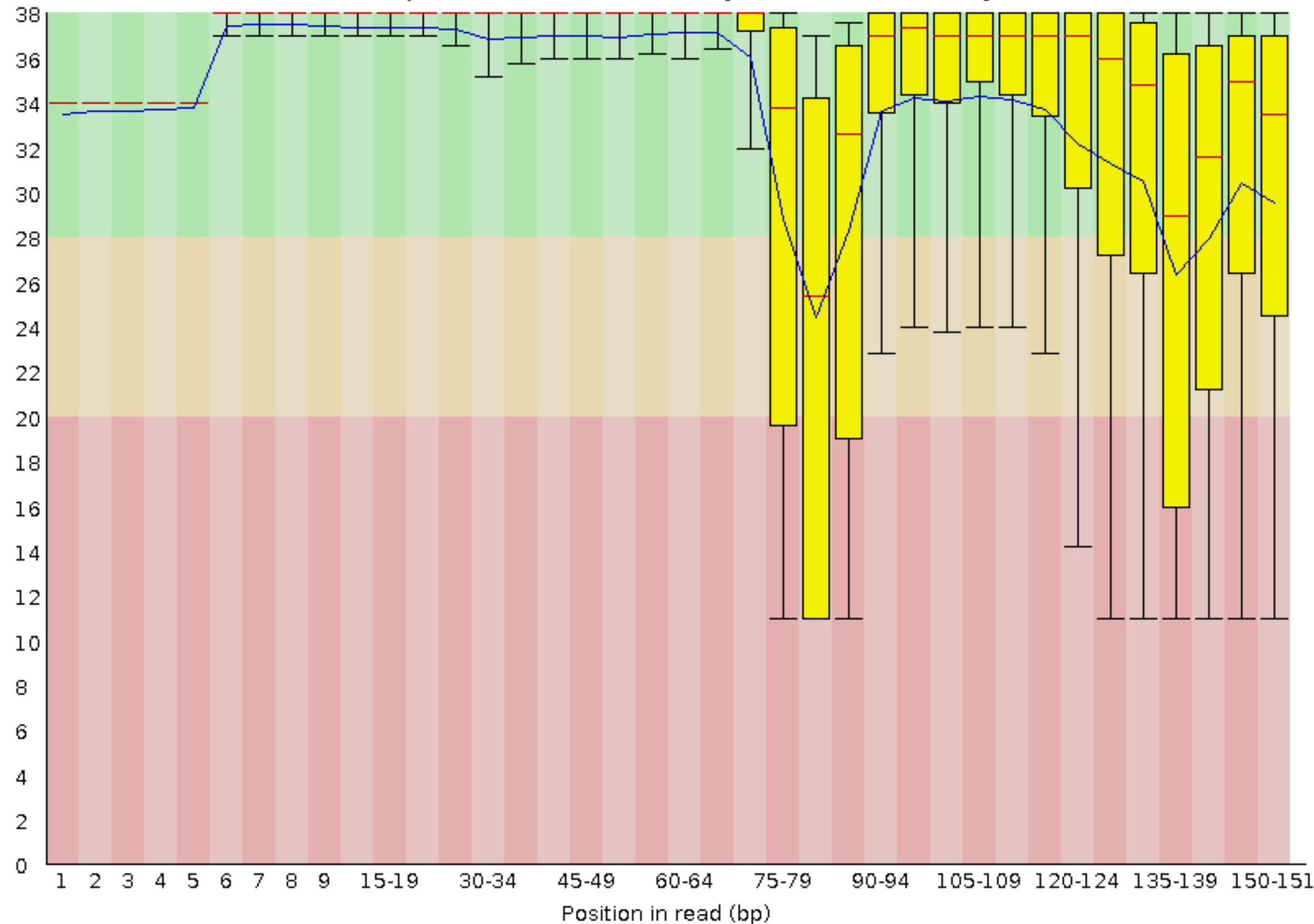
Per base sequence quality

Quality scores across all bases (Sanger / Illumina 1.9 encoding)



good sequencing quality

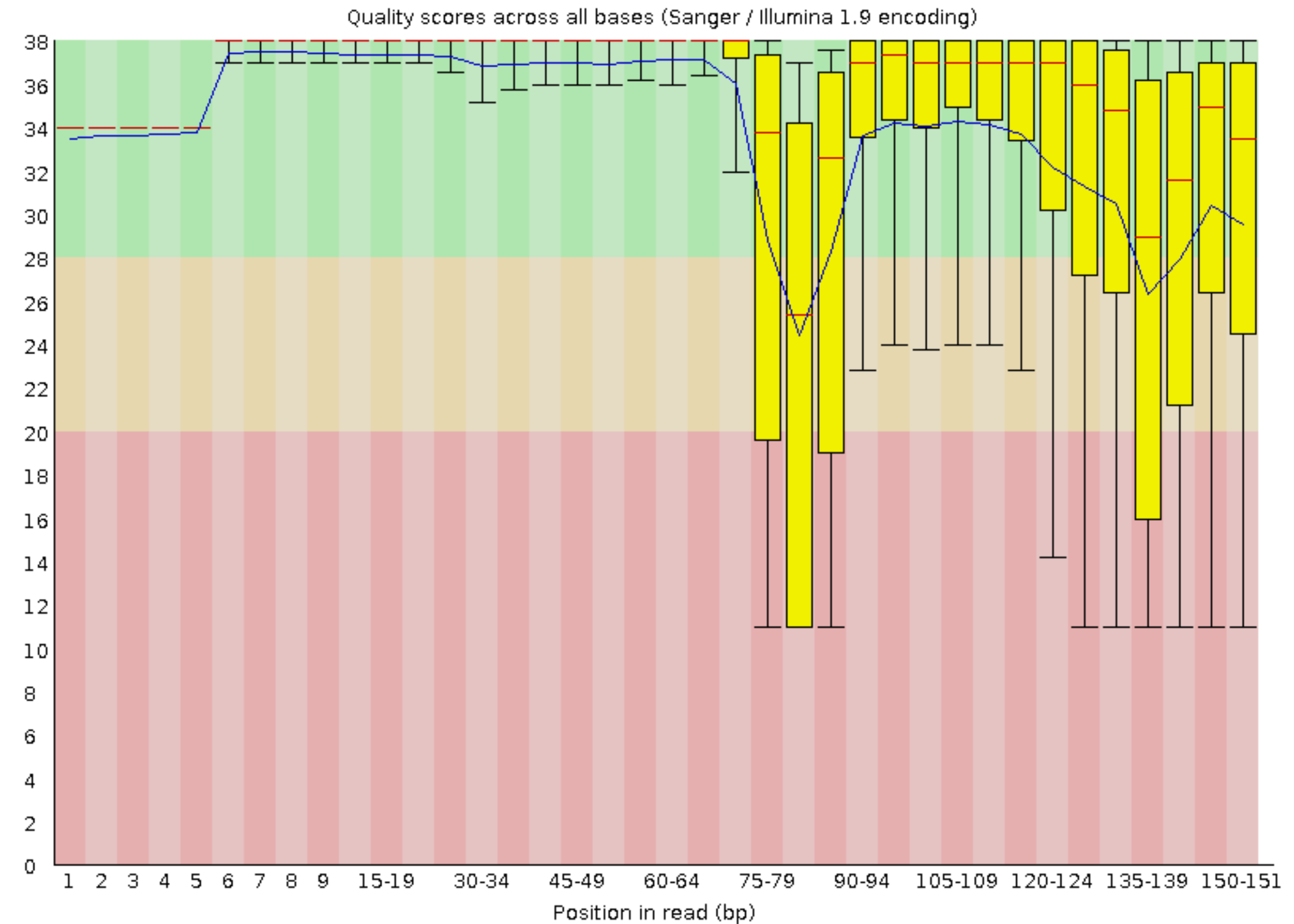
Quality scores across all bases (Sanger / Illumina 1.9 encoding)



poor sequencing quality

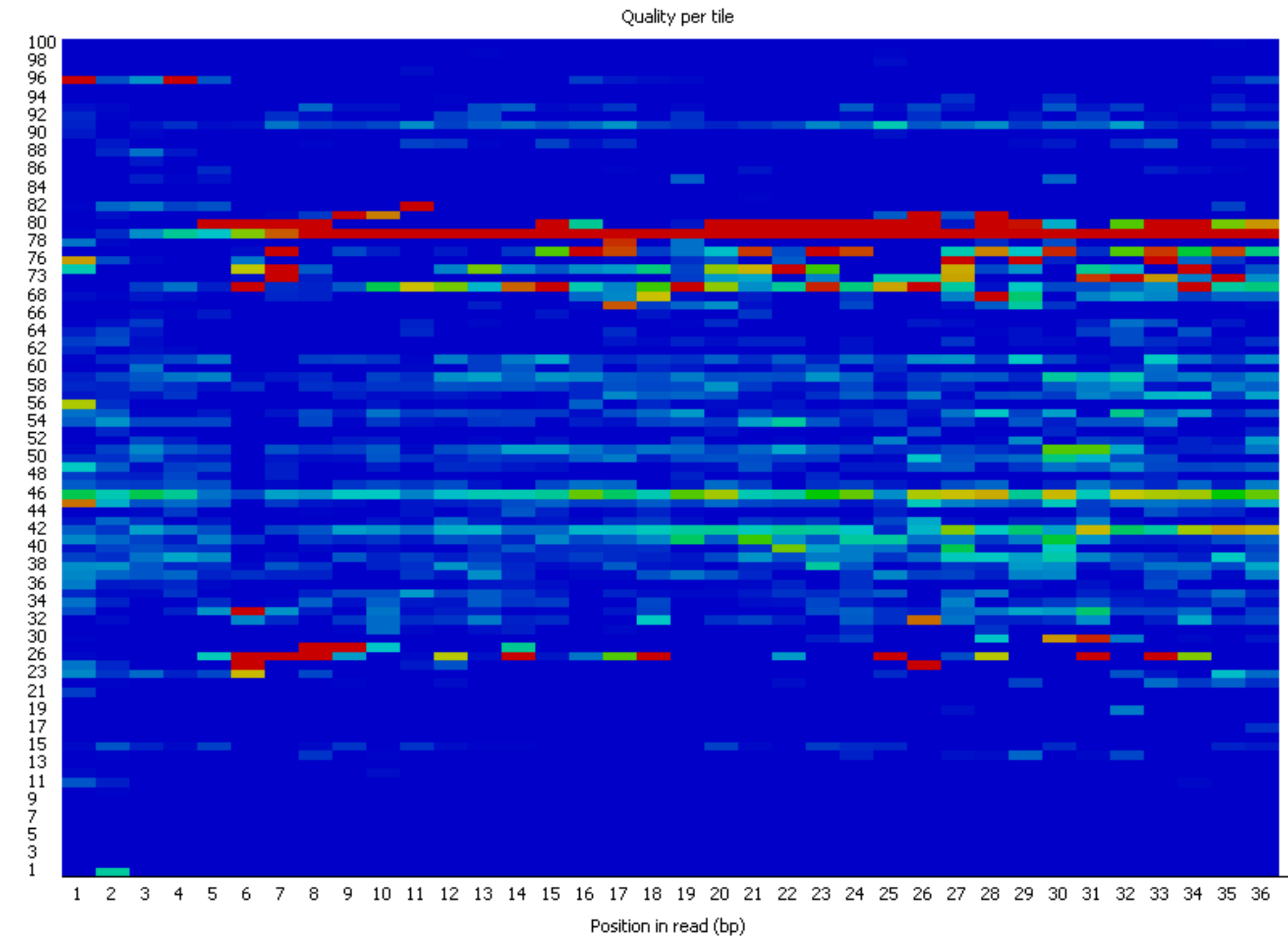
Per base sequence quality

- overview of the range of quality values (y-axis) across all bases at each position (x-axis)
- red line = median value
- blue line = mean value
- yellow box = inter-quartile range (25-75%)
- upper and lower whiskers = 10% and 90% points

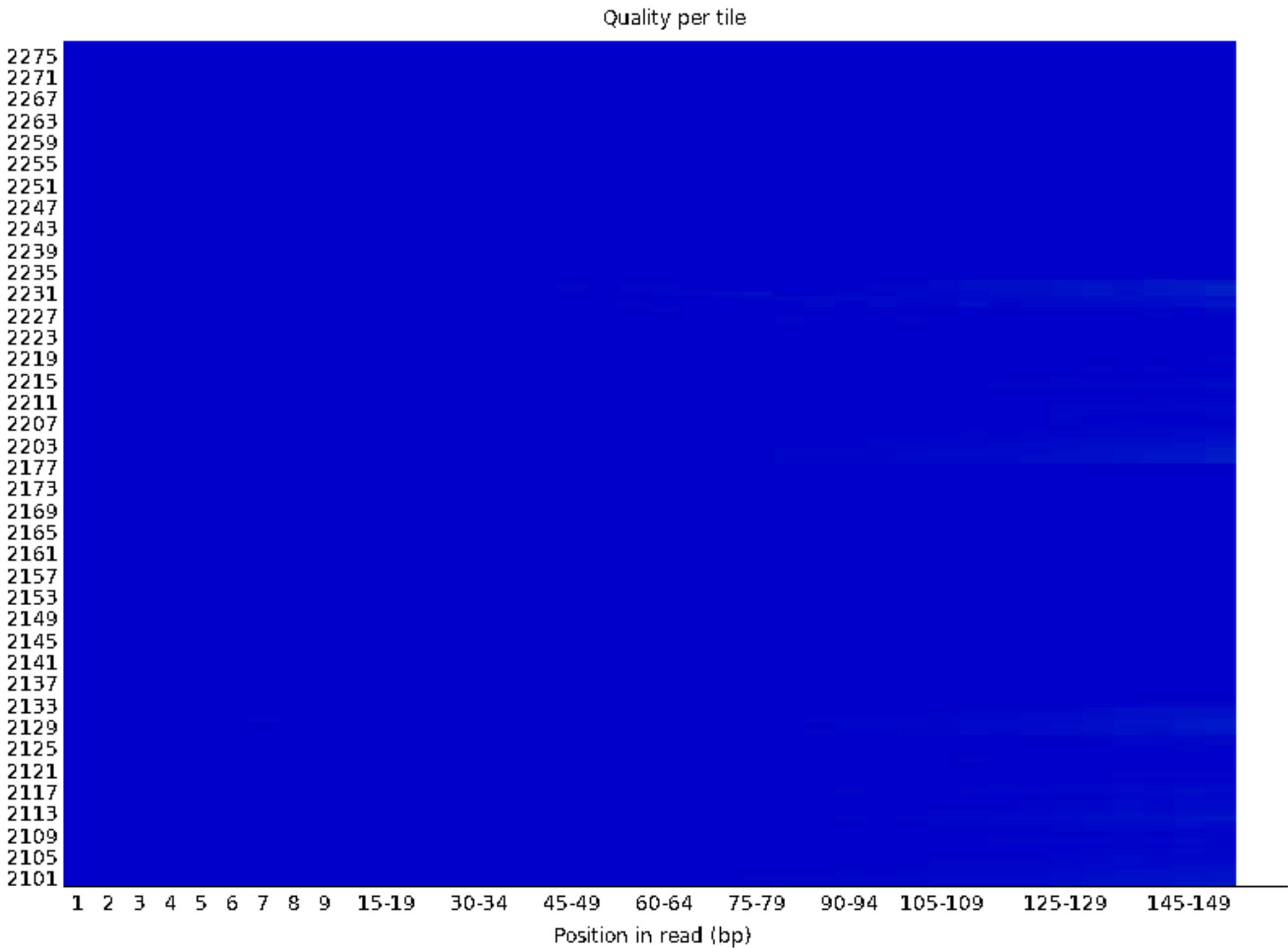


Per tile sequence quality

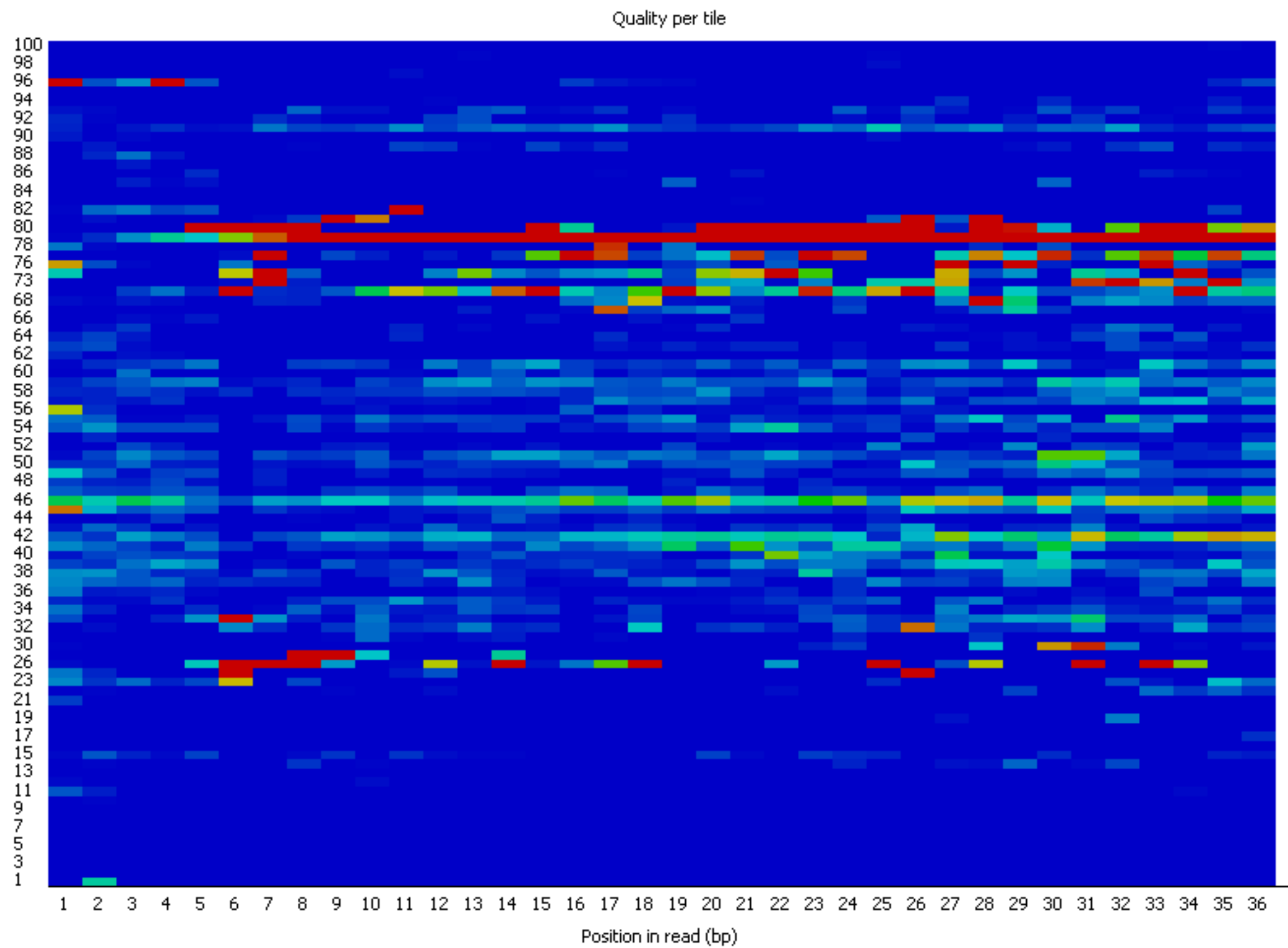
- the deviation from the average quality for each tile
- the sequence identifier of the reads encode from which tile of the flowcell the read comes from
- a loss in quality associated with only one part of the flowcell due to air bubbles, or more permanent problems such as smudges on the flowcell or debris inside the flowcell lane.



Per tile sequence quality

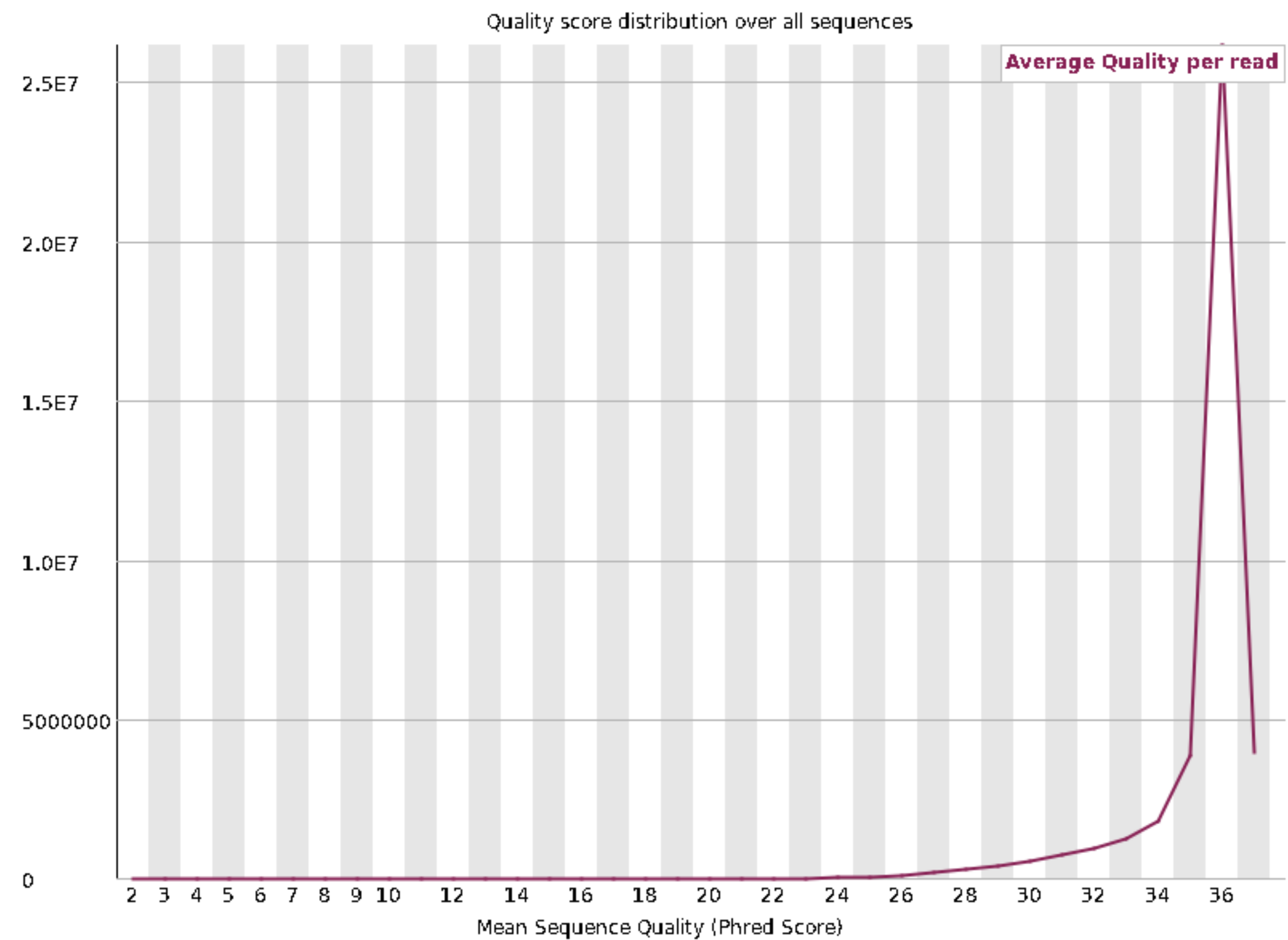


good sequencing quality



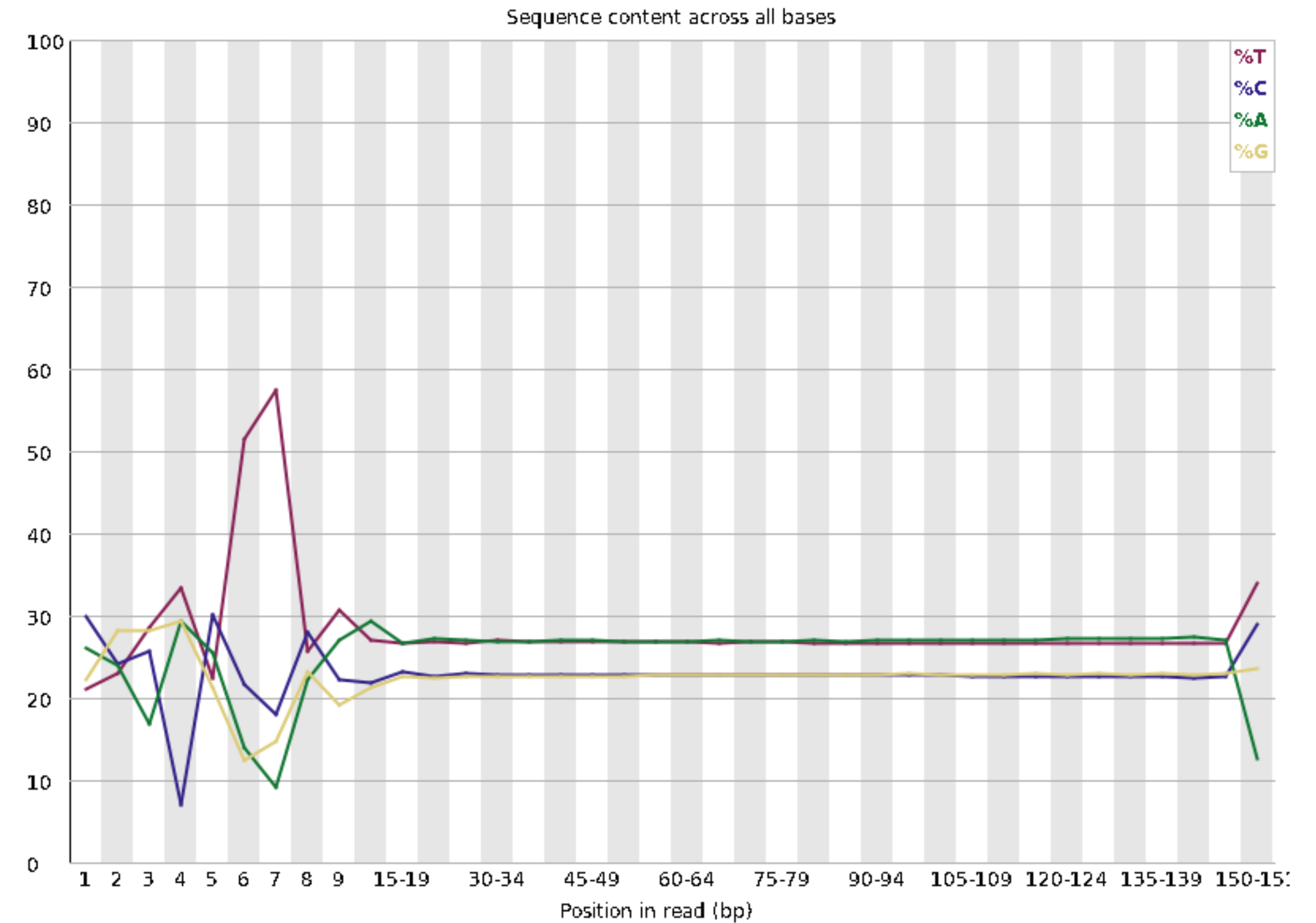
poor sequencing quality

Per sequence quality score



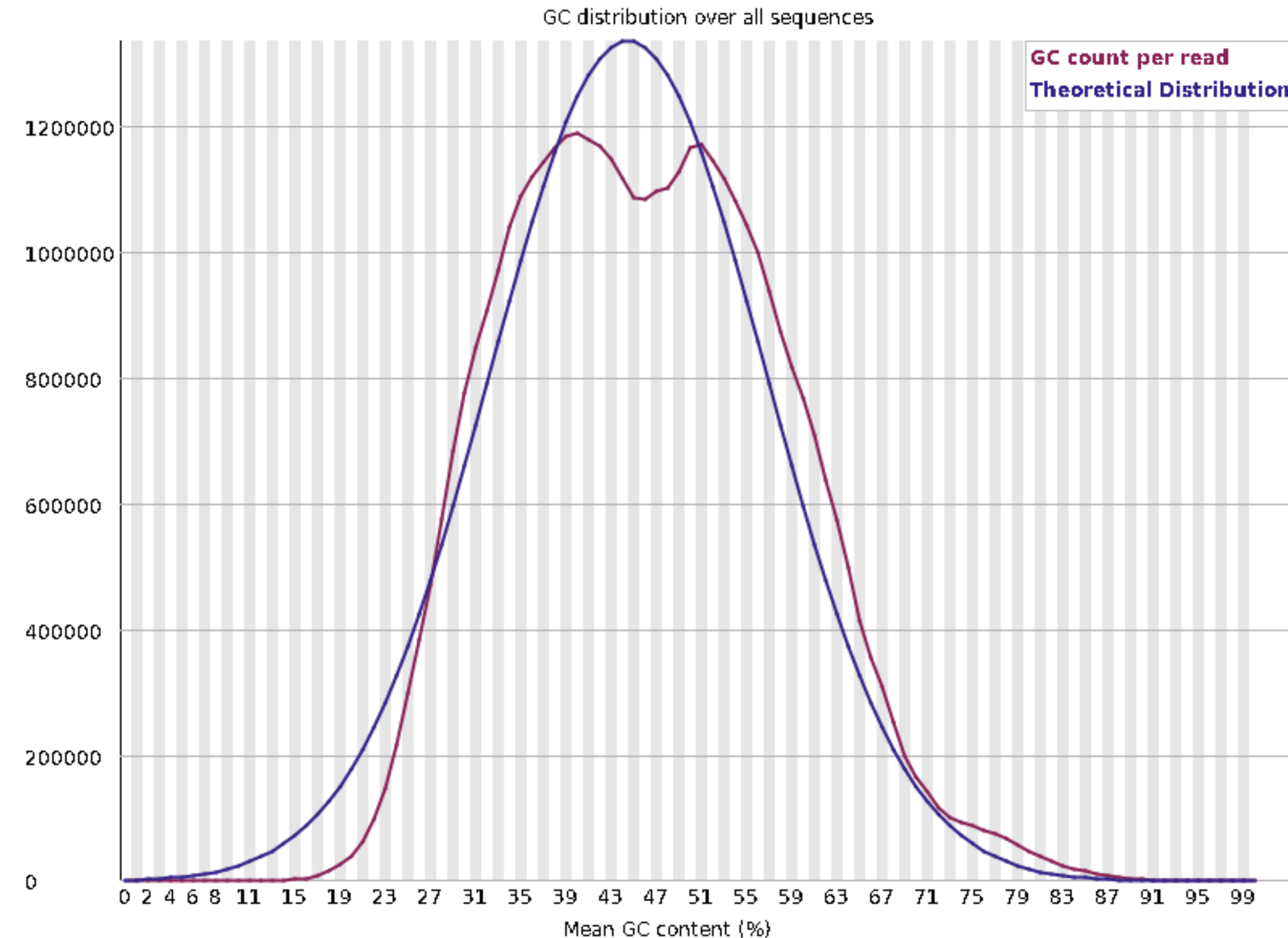
Per base sequence content

- plot of the proportion of each base position
- In a random library there would be little to no difference between the different bases of a sequence run, so the lines in this plot should run parallel with each other
- some types of library will always produce biased sequence composition



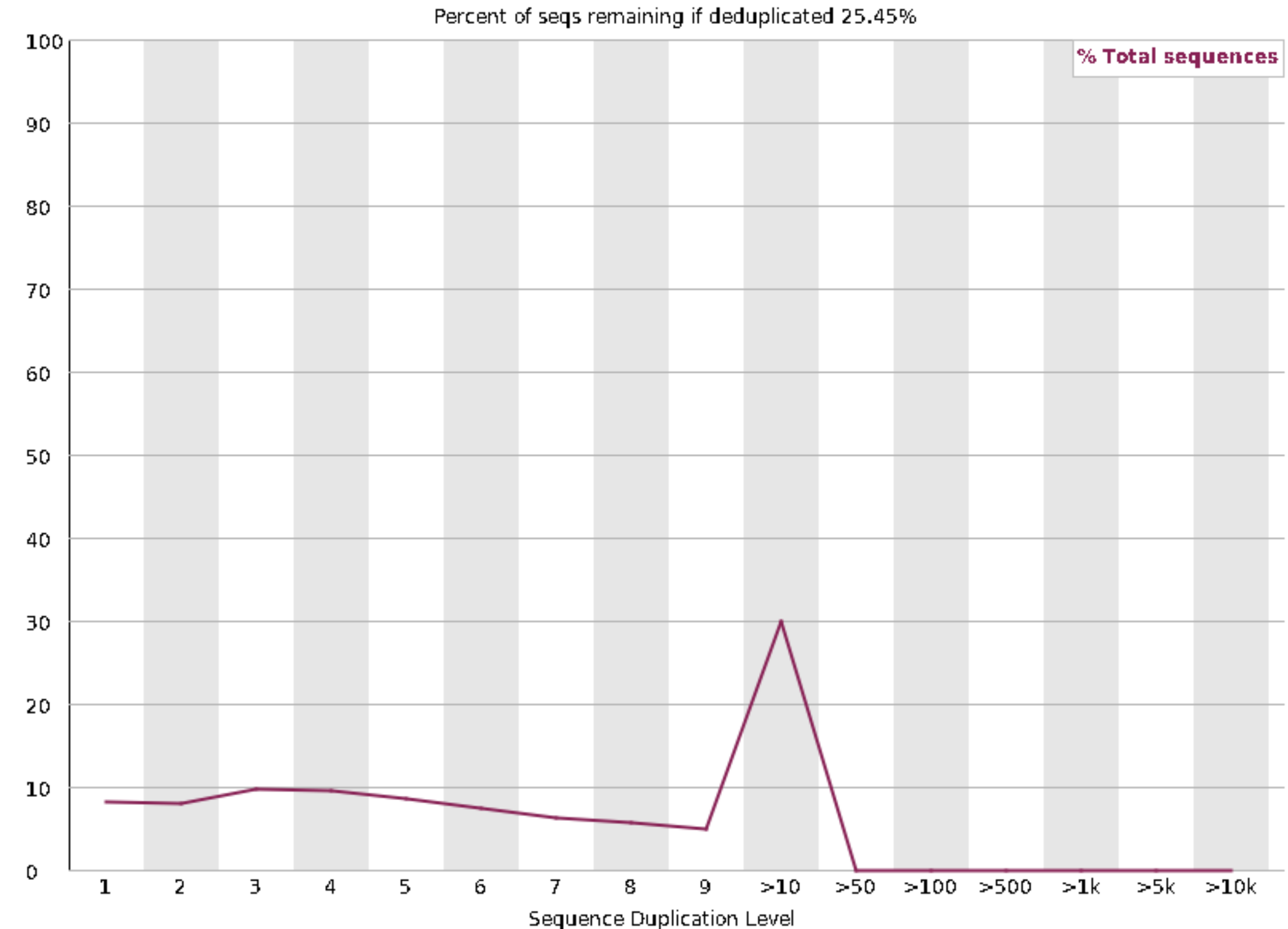
Per sequence GC content

- measures the GC content across the whole length of each sequence in a file and compares it to a modelled normal distribution of GC content.
- a great indicator for:
 - contamination
 - various types of sequence bias (rRNA, high duplication rates, etc.)
- shown plot is from a targeted enrichment sample



Sequence duplication levels

- plot of the distribution of the degrees of duplication of each sequence in the file.
- perfect scenario: a diverse library in which most sequences will occur only once.
- a low level of duplication may indicate a very high level of coverage of the target sequence
- a high level of duplication is more likely to indicate some kind of enrichment bias (eg PCR over amplification).



Quality control - Exercise

Let us review a fastQC report together!

- Check the report J0001_S1_R1_001_fastqc.html from the training material
- What additional information is present in the report that we haven't discussed already?

Conda / Bioconda/Anaconda/ Miniconda

Conda - platform- and language-independent package manager for **easy installation and version management of software**

Bioconda - conda channel, that contains the names, locations and dependencies of many bioinformatics tools

Anaconda - an installable bundle that includes python, the most common data science packages for python (including things like scipy, numpy, pandas and matplotlib) and the conda package manager; also a conda channel (list of packages incl. dependencies),

Miniconda -minimal installer of conda; contains also python, but only few additional packages

Practical Exercise: Data QC

1. Install miniconda. Check <https://docs.conda.io/projects/miniconda/en/latest/> and run the appropriate “quick command line install”
2. Create a new environment named NGS. Use the conda cheat sheet in the training materials.
3. Is fastQC included in Bioconda? If so, please install fastQC using conda in the NGS environment.
4. Download a test datafile from [here](#)
5. Run fastQC on the file from command line. Please check the fastQC documentation for input parameters.
6. If you are ready, please check out the tool MultiQC.